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Research on Sub-Pixel Accuracy Flange Disk Dimension Measurement Based on Machine Vision

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Abstract

Addressing the inefficiencies and inaccuracies inherent in manual measurements of flange disk dimensions, we propose a machine vision-based method for sub-pixel measurement, alongside a dedicated visual measurement system. This system utilizes a CCD camera, calibrated via PyCharm software, to capture images of the flange disk. These images undergo preprocessing—conversion to grayscale, filtering, noise reduction, and binary segmentation—to delineate the flange disk area for further analysis. For edge detection, an improved Canny adaptive threshold selection algorithm is employed. Additionally, we introduce a new edge judgment criterion based on the Zernike algorithm, which, in conjunction with the advanced Canny operator for initial positioning, allows our refined Zernike matrix algorithm to accurately identify the flange disk edge with sub-pixel precision. The least squares fitting algorithm is then applied to determine the flange disk's outer and inner diameters accurately. A comparative analysis of experimental results demonstrates that our enhanced algorithm significantly reduces relative errors compared to both its predecessor and manual measurements. Importantly, the measurements obtained with the improved algorithm closely match manual measurements and are markedly more accurate than those achieved with the traditional Zernike matrix algorithm. This level of precision ensures that our measurement method meets the high accuracy requirements of industrial parts manufacturing.

Keywords Flange disk · Sub-pixel measurement · PyCharm · Zernike matrix algorithm

1 Introduction

The automotive flange disk is a key component of the vehicle's suspension system, connecting the wheel axle to the suspension. It provides essential support and stability for the wheel axle, enabling smooth wheel rotation while absorbing and transmitting forces and vibrations during operation. However, variations in materials and manufacturing techniques can lead to significant discrepancies in flange disk dimensions, compromising precision. Currently, companies predominantly use manual inspection or traditional measuring instruments to assess flange dimensions. This approach not only is inefficient but also yields a high rate of inaccuracies, inadequately ensuring product quality.

In industrial inspection, machine vision is primarily used

for measuring the dimensions of mechanical components, offering advantages like non-destructive testing, speed, and cost-efficiency. Its use in modern industrial production is growing [1]. For instance, Chao et al. [2] proposed an improved Zernike moment to measure shaft parts by improving the threshold, and its measurement accuracy can meet the production requirements of industrial parts. Similarly, Kawasue et al. [3] developed a robot, utilizing machine vision, to measure sewage pipelines' inner diameters with an average error of 0.5 mm. Furthermore, Leo et al. [4] created a machine vision system for identifying mechanical and electrical components. This system can quantitatively measure component sizes and detect loose wires using strategically placed top and front cameras. Desmond et al. [5] established a machine vision system that achieves sub-pixel precision in measuring industrial gears, reducing calibration errors to 0.06 pixels with a telecentric lens. Kavitha et al. [6] crafted a system for measuring the radial error of a lathe's main shaft, employing the Hough circle transform to identify the shaft's center and Fourier series curve fitting to analyze its periodic and non-periodic movements, achieving sub-micrometer level repeatability in radial error measurement. Xie et al. [7] utilized machine vision for size measurement of hole-type components, employing median filtering to remove salt-and-pepper noise and the Canny operator for edge

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detection. However, errors were produced due to the reflective surface of metal components. Huang et al. [8] explored a combination of stereo vision and dynamic plane fitting for determining hole radius and spacing. They introduced a technique for inner edge extraction of angles via circular band contraction and improved measurement accuracy through adaptive weight optimization in data processing, albeit at increased hardware costs.

The major contribution of this research work is summarized below,

- To enhance precision in visual measurement of flange disks for industrial automation, an improved method for sub-pixel size measurement using the Zernike matrix has been proposed.
- The traditional Canny operator has been enhanced by integrating bicubic filtering in lieu of Gaussian filtering for the noise reduction phase. For the computation of gradient magnitudes, this approach maintains the conventional Sobel operator as its foundation but enriches it by including templates for 45° and 135° directions. During the threshold selection phase, the Otsu method is employed to derive adaptive dual thresholds, facilitating the coarse localization of edge coordinates within a circular aperture. This modification not only refines the accuracy of edge detection but also improves the overall image analysis process.
- Following the initial enhancements, the edges of the target region on the flange disk intended for measurement are identified. The refined Zernike matrix sub-pixel detection algorithm is then applied to accurately pinpoint edge points. This enhanced algorithm utilizes a Zernike matrix parameter template designed for higher detection accuracy and improves upon the traditional criteria for edge determination. Ultimately, the measurement of dimensions is accomplished by applying the least squares method to fit the contours of both the internal and external diameters of the circular hole component, ensuring precise size determination.

The subsequent organization of this paper is as follows: The second section introduces the theoretical basis and

specific enhancements of the improved algorithm. Experimental results and analysis are presented in the third section. Finally, the fourth section provides a comprehensive summary of the entire paper.

2 Theoretical framework

2.1 Zernike moment principle

Its appeal stems from its rotational invariance, low noise sensitivity, and capability to accurately compute key parameters [9]. These parameters include the step grayscale value of the Zernike matrix edge model, the background grayscale value, the vertical distance from the image's center to its edge, and the angle between the X-axis and the line perpendicular to the image edge. The ability to determine sub-pixel coordinates through these parameters significantly enhances the precision of edge detection in images [10].

A continuous image $f(x, y)$ is characterized by its n th order m th Zernike matrix Z_{nm} , as defined by equation (1):

$$Z_{nm} = \frac{n+1}{\pi} \iint_{x^2+y^2 \leq 1} f(x, y) V_{nm}^*(\rho, \theta) d_x d_y \quad (1)$$

where x and y satisfy the condition $x^2 + y^2 \leq 1$; $V_{nm}^*(\rho, \theta)$ represents the complex conjugate of the n th order m th Zernike matrix polynomial, ρ is the vector length from the origin to the point (x, y) , and θ is the angle between this vector and the X-axis in the counterclockwise direction.

Under discrete conditions, the Zernike moments are shown in equation (2):

$$Z_{nm} = \sum_x \sum_y f(x, y) V_{nm}^*(\rho, \theta) \quad (2)$$

2.2 Principle of sub-pixel edge detection by Zernike moment method

The ideal edge step model is illustrated in Fig. 1. (x_0, y_0) represents the origin coordinates; L denotes the ideal image edge; k is the step amplitude between the background and the target area; h is the background grayscale value; l is the vertical distance from the center to L ; and α is the angle between line l_1 and the X-axis.

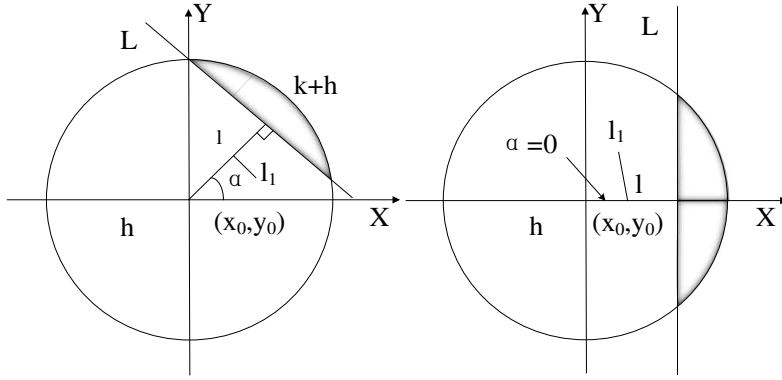


Fig. 1 Ideal edge step model

If the edge is rotated by $-\alpha$, the relationship between the rotated Zernike matrix Z'_{nm} and the original Zernike matrix Z_{nm} is as shown in equation (3):

$$Z'_{nm} = Z_{nm} e^{-pm\alpha} \quad (3)$$

Where: p is an imaginary unit.

Utilizing the imaginary part, $\text{Im}[Z_{11}]$, and the real part, $\text{Re}[Z_{11}]$, of Z_{11} , the rotation angle, denoted as α , can be determined, as depicted in equation (4):

$$\alpha = \arctan \frac{\text{Im}[Z_{11}]}{\text{Re}[Z_{11}]} \quad (4)$$

The derived Z'_{00} , Z'_{11} , and are represented as shown in the equation(5):

$$\begin{cases} Z'_{00} = Z_{00} = h\pi + \frac{k\pi}{2} - k \arcsin(l) - kl\sqrt{1-l^2} \\ Z'_{11} = Z_{11} e^{-p\alpha} = \frac{2k}{3} (1-l^2)^{\frac{3}{2}} \\ Z'_{20} = Z_{20} = \frac{2k}{3} l(1-l^2)^{\frac{3}{2}} \end{cases} \quad (5)$$

The parameters h , k , and l can be derived from the equation, as shown in equation (6):

$$\begin{cases} h = \frac{Z_{00} - \frac{d\pi}{2} + d \arcsin(l) + dl\sqrt{1-l^2}}{\pi} \\ d = \frac{3Z'_{11}}{2(1-l^2)^{\frac{3}{2}}} \\ l = \frac{Z'_{20}}{Z'_{11}} \end{cases} \quad (6)$$

The final sub-pixel edge detection formula is derived, as shown in equation (7):

$$\begin{bmatrix} x_s \\ y_s \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + l \begin{bmatrix} \cos \alpha \\ \sin \alpha \end{bmatrix} \quad (7)$$

Where: (x_s, y_s) are the sub-pixel edge coordinates.

Then the Zernike moment sub-pixel coordinates of the unit circle when sampled in the discrete case using the $N \times N$ template are calculated as shown in equation (8) as follows:

$$\begin{bmatrix} x_s \\ y_s \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \frac{N}{2} \begin{bmatrix} \cos \alpha \\ \sin \alpha \end{bmatrix} \quad (8)$$

2.3 Improved Canny Edge Detection Algorithm

The Canny algorithm stands as a prominent method for edge detection, comprising four main steps: smoothing the image through convolution with a two-dimensional Gaussian filter; calculating the gradient magnitude and direction for every pixel; applying non-maximum suppression to the gradient magnitude to thin edges; identifying and connecting edges via a dual-threshold technique [11]. Despite its widespread use, the traditional Canny algorithm faces several challenges: Gaussian filtering may compromise edge integrity; Gradient calculation based on a 2x2 neighborhood is susceptible to noise, potentially leading to the omission of crucial edge information; The fixed dual-threshold setting lacks flexibility and adaptability to varying image conditions [12,13] .

(1) Bilateral filtering instead of Gaussian filtering

Noise may be introduced during the image acquisition process due to external influences. Given that Gaussian filtering does not adequately preserve edge integrity, this system adopts bilateral filtering for noise reduction to safeguard edge details [14]. Bilateral filtering is an advanced non-linear technique that enhances Gaussian filtering by including pixel value weighting, which considers both spatial

proximity and grayscale value similarity [15]. This approach effectively maintains crucial edge information in the image while simultaneously reducing noise.

The bilateral filtering is shown in equation (9):

$$g(i, j) = \frac{\sum_{k,l} f(k, l) \omega(i, j, k, l)}{\sum_{k,l} \omega(i, j, k, l)} \quad (9)$$

where i and j are the coordinate points of the current convolved pixel and k and l are the coordinate points of the domain pixel, respectively.

(2) Sobel operator to compute gradient magnitude

In edge detection algorithms, gradient magnitude is a key indicator of pixel variation within an image. This paper enhances the traditional Sobel operator by incorporating direction templates for 45° and 135° , expanding the operator's capability to include horizontal, vertical, 45° , and 135° directions. The calculation of first-order partial derivatives now spans an 8-pixel neighborhood across the x, y, 45° , and 135° directions, making the enhanced template more precise in edge localization and superior in noise suppression compared to its traditional counterpart. The specific templates are presented in Equation (10):

$$\begin{cases} H_x = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}, H_y = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \\ H_{45^\circ} = \begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & 1 \\ -2 & -1 & 0 \end{bmatrix}, H_{135^\circ} = \begin{bmatrix} -2 & -1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix} \end{cases} \quad (10)$$

Where H_x , H_y , H_{45° , and H_{135° represent the horizontal, vertical, 45° , and 135° components of the gradient template, respectively.

The original image, once processed through the planar convolution with the multi-directional Canny template, produces gradient components in four distinct directions. This process of gradient calculation is detailed in Equation (11):

$$\begin{cases} P_x(i, j) = g(i+1, j) - g(i-1, j) \\ P_y(i, j) = g(i, j+1) - g(i, j-1) \\ P_{45^\circ}(i, j) = g(i-1, j+1) - g(i+1, j-1) \\ P_{135^\circ}(i, j) = g(i+1, j+1) - g(i-1, j-1) \end{cases} \quad (11)$$

$$\begin{cases} f_x(i, j) = P_x(i, j) + \frac{P_{45^\circ}(i, j) + P_{135^\circ}(i, j)}{2} \\ f_y(i, j) = P_y(i, j) + \frac{P_{45^\circ}(i, j) - P_{135^\circ}(i, j)}{2} \\ M(i, j) = \sqrt{\alpha \cdot f_x(i, j)^2 + \beta \cdot f_y(i, j)^2} \\ M = \sqrt{M_x^2 + M_y^2 + M_{45^\circ}^2 + M_{135^\circ}^2} \end{cases} \quad (12)$$

Equation (12) represents the partial derivatives of the gradient template across various directions, where $f_x(i, j)$ and $f_y(i, j)$ represent the horizontal and vertical directions. The gradient weighting coefficients are denoted by α and β , with the gradient components on each direction being M_x , M_y , M_{45° , and M_{135° . Ultimately, the final gradient M can be obtained.

(3) Non-extreme value suppression

Edges derived purely from gradient values tend to appear blurry. Non-maximum suppression serves to sharpen these edges by comparing the gradient intensity of a given pixel with the gradient values along the direction of the gradient on either side [16]. A pixel is preserved as an edge point, if it represents a maximum value in this comparison; otherwise, it is classified as a non-edge point.

This paper introduces a refined approach to determining edge pixels by not only evaluating the gradient magnitude but also considering the gradient's direction. For instance, if the gradient magnitude in the x-direction exceeds that in the y-direction, the decision to classify a pixel as an edge is based on the x-direction gradient. Conversely, if the y-direction gradient is greater, the decision relies on the y-direction. By incorporating the direction of the gradient into our analysis, we enhance our ability to accurately detect edges, thereby improving the overall precision of edge detection.

(4) Otsu threshold selection

Following the application of non-maximum suppression, residual noise and spurious edges may persist in the image [17]. To mitigate these, the Canny algorithm employs a dual-threshold technique aimed at minimizing the occurrence of false edges [18]. To address the inherent difficulties in choosing appropriate dual thresholds, this work enhances the traditional Canny threshold selection process through the adoption of the Otsu method. This approach allows for the automatic determination of adaptive dual thresholds, significantly refining the process beyond the constraints of manual, experience-based threshold setting. The details of this improved algorithm are as follows:

Assuming there exists a threshold value Th that divides all pixels in the image into two classes, C_1 and C_2 , with C_1 and C_2 having grayscale means m_1 and m_2 , respectively, and the

global average grayscale value of the image being m . The probabilities of a pixel being classified into C_1 and C_2 are p_1 and p_2 , respectively, as shown in Equation (13):

$$\begin{cases} p_1 \times m_1 + p_2 \times m_2 = m \\ p_1 + p_2 = 1 \\ \sigma^2 = p_1(m_1 - m)^2 + p_2(m_2 - m)^2 \end{cases} \quad (13)$$

where σ^2 is at its maximum, the optimal threshold T_h is selected as the high threshold for the dual threshold, and $0.5 \times T_h$ is chosen as the low threshold for the dual threshold.

2.4 Improved Canny-Zernike moment sub-pixel edge detection algorithm

The conventional approach calculates Zernike moments by taking the image's center of mass as the origin of coordinates and projecting the image onto a unit circle. In contrast, the refined Canny-Zernike sub-pixel edge detection algorithm eliminates the need to scan the entire image. As a result, leveraging the sub-pixel edge detection method significantly accelerates the detection process.

(1) Selection of Zernike template coefficients

The choice of Zernike template orders directly impacts the precision of image edge detection at the subpixel level. Typically, selecting a larger template size leads to higher detection accuracy but also requires more time for processing. Given the objectives of this study, a 7×7 Zernike matrix parameter template has been chosen to improve the algorithm's detection accuracy further.

(2) Improvement of edge judgment conditions of Zernike moment algorithm

The enhanced algorithm expands upon Ghosal's method by extending the coefficient templates from Z_{00} , Z_{11} , Z_{20} to Z_{31} , Z_{40} . Concurrently, it introduces a new criterion for edge determination, $k \geq k_i \cap |l_2 - l_1| \leq l_i$, replacing the conventional condition of $k \geq k_i \cap l_2 - l_1 \leq l_i$. The specifics of this algorithm are detailed in Equation (14):

$$\begin{cases} Z'_{31} = K \left[\frac{4}{5} l^2 (1 - l^2)^{\frac{3}{2}} - \frac{2}{15} (1 - l^2)^{\frac{3}{2}} \right] \\ Z'_{40} = K \left[-\frac{2}{5} l^2 (1 - l^2)^{\frac{3}{2}} + \frac{16}{15} l^3 (1 - l^2)^{\frac{3}{2}} \right] \end{cases} \quad (14)$$

Derived from equation (5) and equation (15):

$$\begin{cases} l_1 = \sqrt{\frac{5Z'_{40} + 3Z'_{20}}{8Z'_{20}}} \\ l_2 = \sqrt{\frac{5Z'_{31} + 3Z'_{11}}{6Z'_{11}}} \\ k = \frac{3Z'_{11}}{2(1 - l_2^2)^{\frac{3}{2}}} \end{cases} \quad (15)$$

The specific implementation steps of the improved algorithm are as follows:

Step 1: The traditional Zernike moments detection algorithm is used to find the Zernike moments of five different orders Z_{00} , Z_{11} , Z_{20} , Z_{31} and Z_{40} ;

Step 2: By the property of rotational variance of the Zernike orthogonal moments, Z'_{00} , Z'_{11} , Z'_{20} , Z'_{31} , Z'_{40} are computed;

Step 3: Derive l_1 , l_2 , and k from Z'_{00} , Z'_{11} , Z'_{20} , Z'_{31} , and Z'_{40} , and compute l ;

Step 4: Calculate k by substituting l into the formula for k and h ;

Step 5: Determine whether each edge pixel point parameter satisfies the decision condition, $k \geq k_i$ and $l \leq 1/\sqrt{2}$;

Step6: The new sub-pixel edge judgment basis $k \geq k_i \cap |l_2 - l_1| \leq l_i$ is used, and then the pixel-level rough edge points obtained using the improved Canny algorithm are discriminated. Those that conform are edge pixel points, and vice versa are not.

3 Experimentation and Analysis

3.1 Camera calibration

For size measurement, it is essential to convert the measured pixel dimensions into the actual dimensions of the workpiece [19]. This requires establishing a conversion relationship between pixel dimensions and actual dimensions [20]. To guarantee the accuracy of measurement results, a 6mm x 6mm black-and-white chessboard calibration board is utilized for camera calibration. The camera captures 20 images of the calibration board from various angles within the measurement area. In each image, two corner points are identified, and both the actual and pixel distances between these points are calculated. The calibration equation is detailed in Equation (16):

$$k = \frac{L}{1} \quad (16)$$

where L is the actual size between two points of the

calibration plate; l is the pixel size between two points of the calibration plate.

The average of 20 sets of K values was obtained, which is the calibration coefficient K . The practical significance is that the physical size of a pixel is 0.08421 mm/pixel.

3.2 Measurement system and inspection process

The hardware components of the detection system mainly include a computer, an industrial camera, a lens, a backlight source, a ring light source, and a test stand [21]. The process involves the industrial camera capturing images of the flange end face, which are then transmitted to the computer for processing. The computer analyzes these images and presents the measurement outcomes. Fig. 2 illustrates the detection system, while Table 1 provides a detailed breakdown of the hardware components.

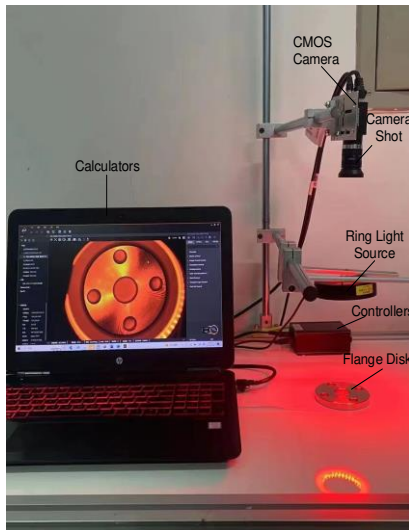


Fig. 2 Machine vision based dimensional inspection device

Table 1 Hardware composition of machine vision-based dimensional inspection device

Hardware composition	Model number
Industrial camera	MV-CS050
Camera shot	MVL-MF0824M
Ring Light Source	YS-R90-60-20
Calculators	Intel(R) Core (TM) i5-7300HQ

3.3 Image Preprocessing

3.3.1 Grayscale

Given that the camera captures color images across three channels, which encompass an extensive array of features and computational demands, direct analysis can be notably time-intensive [22]. Converting these images to grayscale simplifies them to a single-channel format, significantly enhancing computational efficiency [23]. For this conversion,

a weighted average method is employed. The effectiveness of this approach is illustrated through the comparison of original and grayscale images of the flange's outer circle and inner hole, as depicted in Fig. 3:

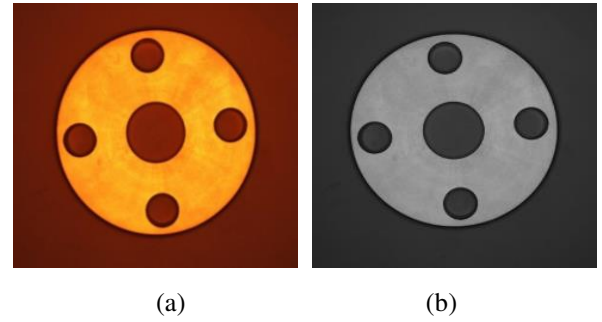
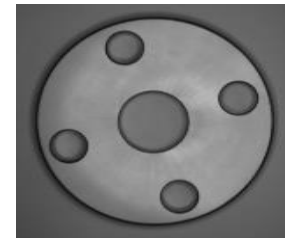


Fig. 3 Original and grey scale images of flange outer circle and bore images: **a** Original image and **b** Grey scale image

3.3.2 Filtering and denoising

The generation of the original image is often marred by noise, which compromises image quality and obscures details. This, in turn, adversely affects the efficacy of image segmentation and edge detection. Consequently, it is imperative to employ suitable techniques to minimize noise and thereby enhance image quality [24]. Among various noise reduction methods, median filtering stands out for its ability to preserve image edge information while effectively eliminating unwanted noise [25], as illustrated in Fig.4.



Figs.4 Comparison of noise reduction effect of different filters

3.3 Image Segmentation

Employing image segmentation to isolate regions of interest can streamline the subsequent image processing steps. Threshold-based segmentation methods are particularly effective in scenarios where there is a stark contrast between the foreground and background [26]. In this study, the contrast between the flange and its background is pronounced, making the threshold-based segmentation approach well-suited for segmenting the image. This method involves binarization segmentation, where a specific threshold is predetermined for the segmentation process.

The histogram is used to show the gray level distribution of the image, according to the gray level histogram shows that the segmentation effect is better when $T=88$ is selected. The

binarized segmentation graph is shown in Figs. 5:

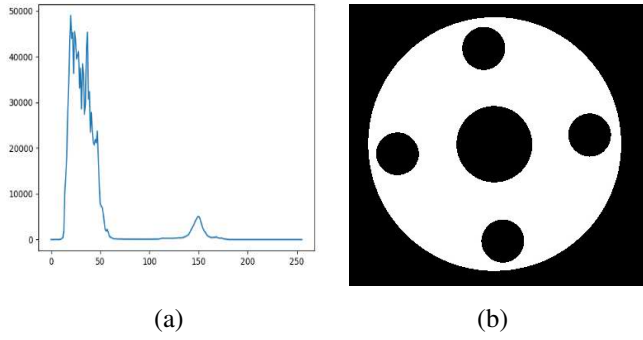


Fig.5 Binary segmentation map: **a** Grey scale histogram and **b** Segmentation plot

3.4 Algorithm Flowchart

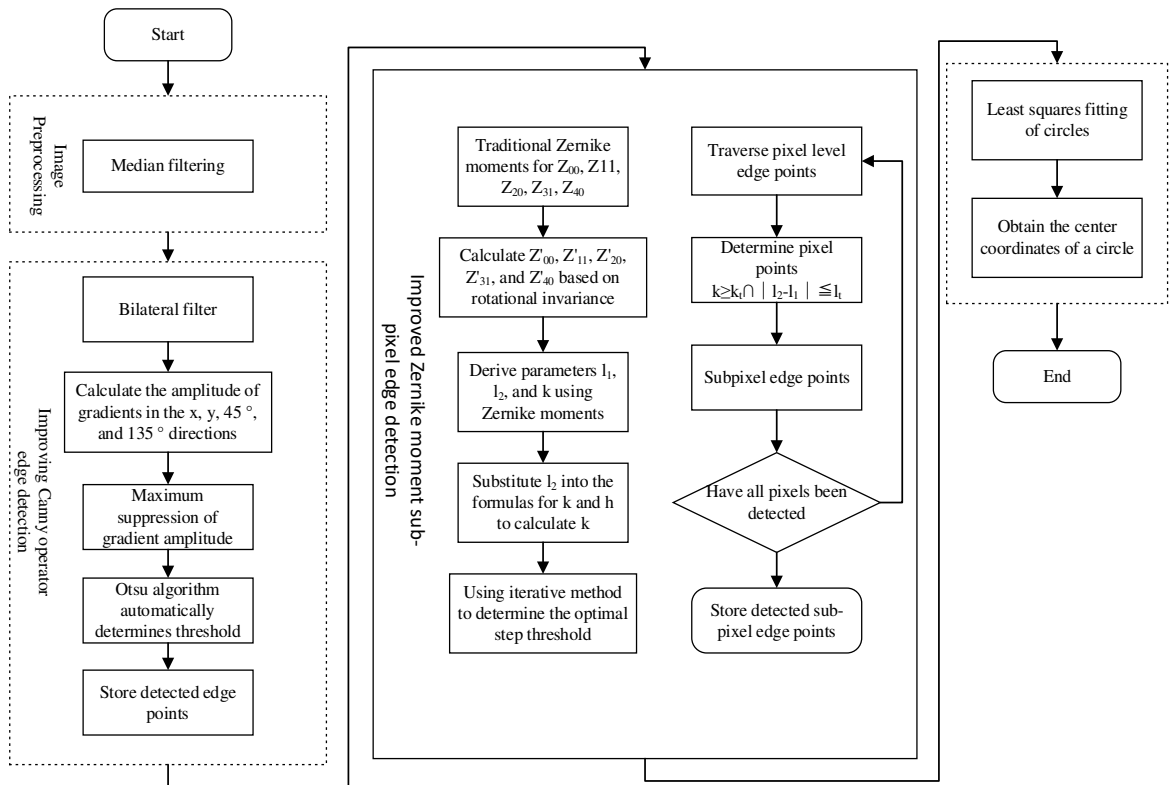


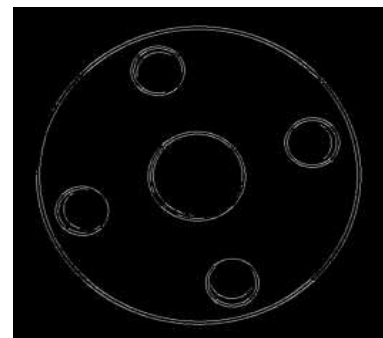
Fig. 7 algorithm flow chart

3.5 Analysis of results

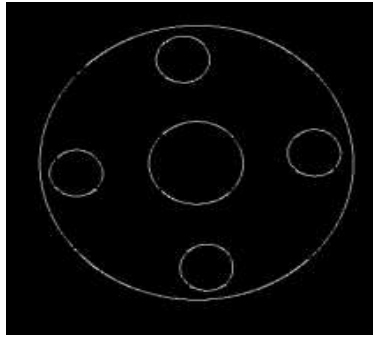
To conduct a qualitative comparison between traditional and enhanced algorithms, both the conventional and the improved Zernike algorithms were employed to extract subpixel edge points of the inner and outer contours of a part, as shown in Fig.8. Fig.8a displays the edge extraction performed by the traditional algorithm on the flange, while Fig.8b illustrates the results achieved with the improved algorithm. From these observations, it is evident that the traditional algorithm only provides a vague detection of the flange's contour edges, missing critical point information, exhibiting more edge noise,

This paper proposes a sub-pixel detection algorithm for the round holes in parts, utilizing an enhanced Zernike moment sub-pixel edge model. Initially, the image undergoes preprocessing, followed by the application of an improved Canny operator for a coarse pixel-level localization of the round hole's edge, yielding pixel-level edge points. Subsequently, a more precise localization is performed using the enhanced Zernike moment sub-pixel edge model, which provides accurate sub-pixel coordinates for the edge points. The final step involves fitting these sub-pixel edge coordinates to the target body, based on the least squares principle, to determine the center coordinates and radius of the round hole. The methodical steps are illustrated in Fig. 7.

and showing some discontinuities in contour points. In contrast, the improved algorithm demonstrates superior accuracy in size measurement and higher precision in edge extraction.



(a)



(b)

Fig. 8 Circular hole edge extraction map: **a** Traditional algorithm edge extraction graph and **b** Edge extraction graph of this paper algorithm.

The study utilizes both traditional and improved algorithms to measure the inner and outer diameters of a flange, with manual measurements serving as a reference point for data comparison. The measurement outcomes for the flange's diameters obtained through the traditional Zernike matrix algorithm and the enhanced algorithm presented in this research, alongside their comparison to manual measurement

values, are detailed in Tables 2-7. Analysis of the data from these tables indicates that the measurements derived from the Zernike matrix subpixel detection algorithm are more precise than those obtained manually. Furthermore, when the results of the improved algorithm are compared to manual measurements, the relative error of the improved algorithm is found to be significantly lower than that of the traditional algorithm. This indicates that the measurements taken using the improved algorithm more closely align with manual measurements, showcasing a higher level of precision over the traditional Zernike matrix algorithm. The introduction of new edge detection criteria and an optimized threshold value selection method by the improved algorithm, as opposed to the conventional criteria and manual threshold adjustment method, results in enhanced precision and efficiency. This advancement lays a robust theoretical groundwork for the application of these measurement algorithms in practical scenarios.

Table 2 Measurement results of flange outer diameter

Times	Anthropometry (mm)	Traditional Zernike Moment Algorithm		Average Relative error %	Improvement of Zernike Moment Algorithm		Average relative error %
		Measurement results	Absolute error		Measurement Results	Absolute error	
		(mm)	(mm)		(mm)	(mm)	
1	80.64	80.528	-0.112	0.1813	80.676	0.036	0.03373
2	80.64	80.783	0.143		80.651	0.011	
3	80.64	80.474	-0.166		80.572	-0.068	
4	80.64	80.516	-0.124		80.631	-0.009	
5	80.64	80.826	0.186		80.652	0.012	

Table 3 Measurement results of flange center hole diameter

Times	Anthropometry (mm)	Traditional Zernike Moment Algorithm		Average Relative error %	Improvement of Zernike Moment Algorithm		Average relative error %
		Measurement results	Absolute error		Measurement Results	Absolute error	
		(mm)	(mm)		(mm)	(mm)	
1	25.04	24.901	-0.103	0.5911	25.019	-0.021	0.1246
2	25.04	24.867	-0.137		25.065	0.025	
3	25.04	25.164	0.160		24.994	-0.046	
4	25.04	25.137	0.133		25.101	0.061	
5	25.04	25.211	0.207		24.901	-0.003	

Table 4 Measurement results of flange bore 1 diameter

Times	Anthropometry (mm)	Traditional Zernike Moment Algorithm		Average Relative error %	Improvement of Zernike Moment Algorithm		Average relative error %
		Measurement results	Absolute error		Measurement Results	Absolute error	
		(mm)	(mm)		(mm)	(mm)	
1	13.25	13.102	-0.148	1.0113	13.287	0.037	0.2294
2	13.25	13.154	-0.096		13.206	-0.044	
3	13.25	13.119	-0.131		13.274	0.024	
4	13.25	13.448	0.198		13.257	0.007	
5	13.25	13.347	0.097		13.210	-0.04	

Table 5 Measurement results of flange bore 2 diameter

Times	Anthropometry (mm)	Traditional Zernike Moment Algorithm		Average Relative error %	Improvement of Zernike Moment Algorithm		Average relative error %
		Measurement results	Absolute error		Measurement Results	Absolute error	
		(mm)	(mm)		(mm)	(mm)	
1	13.20	13.372	0.172	1.1318	13.210	0.01	0.4061
2	13.20	13.080	-0.120		13.119	-0.081	
3	13.20	13.359	0.159		13.154	-0.046	
4	13.20	13.002	-0.198		13.287	0.087	
5	13.20	13.102	-0.098		13.244	0.044	

Table 6 Measurement results of flange bore 3 diameter

Times	Anthropometry (mm)	Traditional Zernike Moment Algorithm		Average Relative error %	Improvement of Zernike Moment Algorithm		Average relative error %
		Measurement results	Absolute error		Measurement Results	Absolute error	
		(mm)	(mm)		(mm)	(mm)	
1	13.25	13.119	-0.131	0.9660	13.338	0.088	0.2491
2	13.25	13.372	0.122		13.274	0.024	
3	13.25	13.415	0.165		13.257	0.007	
4	13.25	13.137	-0.113		13.210	-0.04	
5	13.25	13.359	0.109		13.244	-0.006	

Table 7 Measurement results of flange bore 4 diameter

Times	Anthropometry (mm)	Traditional Zernike Moment Algorithm		Average Relative error %	Improvement of Zernike Moment Algorithm		Average relative error %
		Measurement results	Absolute error		Measurement Results	Absolute error	
		(mm)	(mm)		(mm)	(mm)	

1	13.36	13.474	0.114		13.415	0.055	
2	13.36	13.578	0.218		13.359	-0.001	
3	13.36	13.206	-0.154	1.3653	13.372	0.012	0.2440
4	13.36	13.102	-0.258		13.338	-0.022	
5	13.36	13.528	0.168		13.287	-0.073	

4 Conclusion

This paper tackles the challenge of attaining high precision in the visual measurement of flanges for industrial automation, proposing an enhanced subpixel size measurement method leveraging the Zernike matrix. Initial enhancements focus on improving the noise reduction process of the traditional Canny operator, optimizing the calculation of gradient magnitudes, and refining the threshold selection method, which collectively facilitate the coarse localization of pixel-level edge coordinates at the circular hole's center. Subsequently, edge pixel points within the specified region of the component under inspection are extracted, and the advanced Zernike matrix subpixel detection algorithm is employed for the precise localization of edge points. This improved algorithm utilizes a Zernike matrix parameter template that offers greater detection accuracy and refines the edge detection condition selection method, ultimately allowing for the measurement of the inner and outer diameters of circular hole components via least squares fitting. The algorithm presented in this paper showcases enhanced adaptability and stability in edge detection conditions compared to existing methodologies, thereby boosting detection efficiency and measurement accuracy. Experimental findings confirm that both the relative and absolute errors of measurements obtained with this algorithm are lower than those yielded by current algorithms, satisfying the stringent precision requirements of contemporary precision parts production processes.

Author contributions

Yang Yang proposed an improved method for measuring the inner and outer diameter dimensions of round hole parts and completed the experiments. Guo Jun supervised the writing of the paper and helped to revise and embellish it. Xinyu Chen completed the processing of experimental data and made tables.

Declarations

Conflict of interest The authors declare that there is no conflict of interest.

Ethical approval Not applicable.

Consent for publication Author agreed with the content and that all gave explicit consent to submit.

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